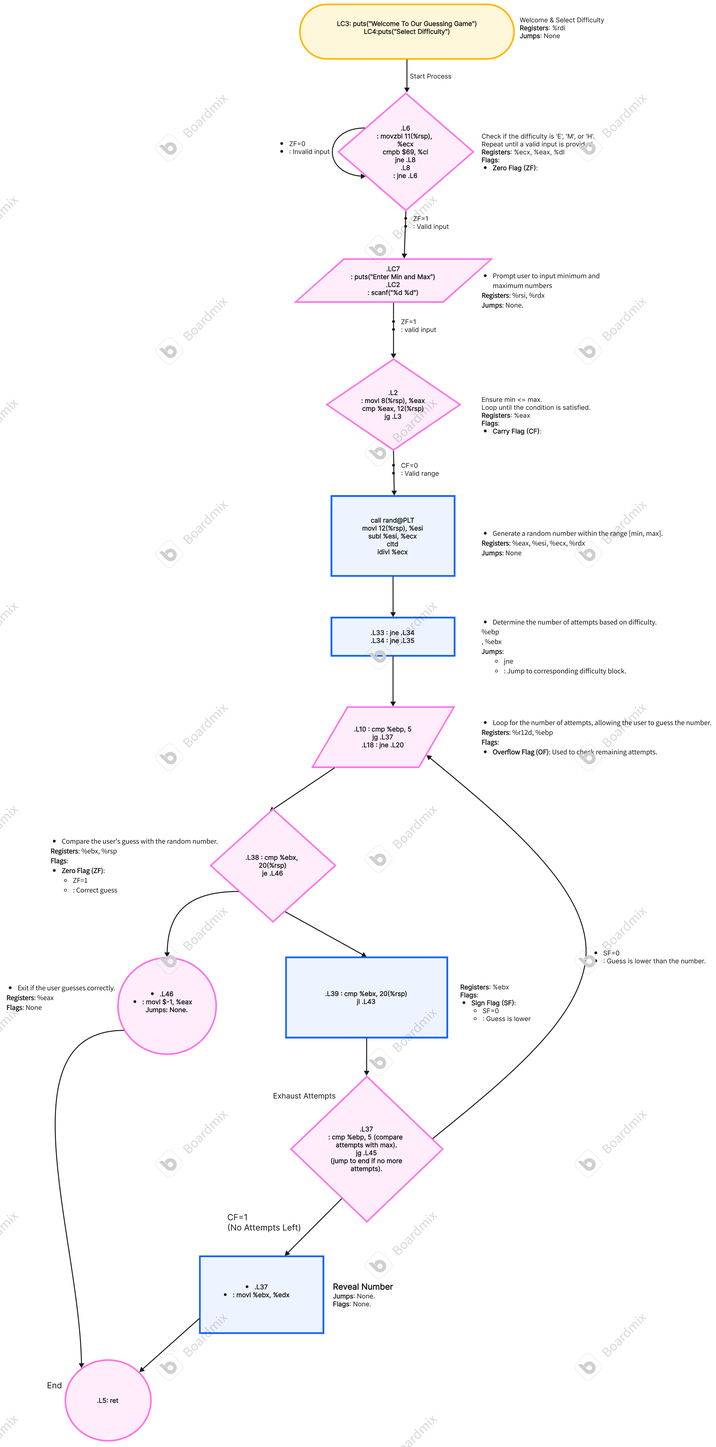
LC3: puts("Welcome To Our Guessing Game")
LC4:puts("Select Difficulty")
Welcome & Select Difficulty
Registers: %rdi
Jumps: None
Start Process
.L6
: movzbl 11(%rsp), %ecx
cmpb $69, %cl
jne .L8
.L8
: jne .L6
ZF=0
: Invalid input
Check if the difficulty is 'E', 'M', or 'H'.
Repeat until a valid input is provided.
Registers: %ecx, %eax, %dl
Flags:
Zero Flag (ZF):
ZF=1
: Valid input
.LC7
: puts("Enter Min and Max")
.LC2
: scanf("%d %d")
Prompt user to input minimum and maximum numbers
Registers: %rsi, %rdx
Jumps: None.
ZF=1
: valid input
.L2
: movl 8(%rsp), %eax
cmp %eax, 12(%rsp)
jg .L3
Ensure min <= max.
Loop until the condition is satisfied.
Registers: %eax
Flags:
Carry Flag (CF):
CF=0
: Valid range
call rand@PLT
movl 12(%rsp), %esi
subl %esi, %ecx
cltd
idivl %ecx
Generate a random number within the range [min, max].
Registers: %eax, %esi, %ecx, %rdx
Jumps: None
.L33 : jne .L34
.L34 : jne .L35
Determine the number of attempts based on difficulty.
%ebp
, %ebx
Jumps:
jne
: Jump to corresponding difficulty block.
.L10 : cmp %ebp, 5
jg .L37
.L18 : jne .L20
Loop for the number of attempts, allowing the user to guess the number.
Registers: %r12d, %ebp
Flags:
Overflow Flag (OF): Used to check remaining attempts.
.L38 : cmp %ebx, 20(%rsp)
je .L46
Compare the user's guess with the random number.
Registers: %ebx, %rsp
Flags:
Zero Flag (ZF):
ZF=1
: Correct guess
SF=0
: Guess is lower than the number.
.L39 : cmp %ebx, 20(%rsp)
jl .L43
Registers: %ebx
Flags:
Sign Flag (SF):
SF=0
: Guess is lower
.L46
: movl $-1, %eax
Jumps: None.
Exit if the user guesses correctly.
Registers: %eax
Flags: None
Exhaust Attempts
.L37
: cmp %ebp, 5 (compare attempts with max).
jg .L45
(jump to end if no more attempts).
CF=1
(No Attempts Left)
.L37
: movl %ebx, %edx
Reveal Number
Jumps: None.
Flags: None.
End
.L5: ret